Pradyun Srivastava

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Education

University of Alberta Class of 2028

Bachelor of Engineering - Computer Engineering (Nanoscale System Design) *Edmonton, AB*

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Databases, Computer Architecture, Data Networks, Machine Learning, Electronic Hardware Design, Digital Logic, Signals & Systems, Circuit Analysis

Experience

Apex Modular Construction May 2026 – August 2026

Software Engineer Intern *Remote*

- Built and prepared a real-time 3D kitchen configurator for launch using React Three Fiber and Three.js, integrating ~30 catalog products and customizable components
- Integrated Gemini AI to translate natural-language commands into configuration changes, enabling users to modify designs optimally
- Developed feasible and dynamic-pricing logic based on cabinet dimensions, wall measurements, ceiling height, opening count, and room complexity

AltHire AI May 2025 – August 2025

Full-Stack Engineer Intern *Remote*

- Built and redesigned user-facing website and career-page features with React and Tailwind CSS, bringing in 50+ applicants
- Tested AI-generated interview assessments against edge cases, analyzing model outputs and collaborating with the product team on identified issues
- Assisted users during platform testing by troubleshooting issues and authored 10+ technical articles covering Python, TypeScript, AI recruiting, and interview preparation

Projects

AlbertaSat Stratospheric Balloon Mission | *Altium, LTSpice, FastAPI, STM32, Arduino* January 2026 – Present

- Leading the Power Team within an interdisciplinary ~20-member high-altitude balloon mission developing systems for launch, tracking, operation, and recovery
- Designing power-distribution PCBs with voltage regulation and overcurrent protection to provide stable power across onboard computing, communications, tracking, sensors, and payload systems
- Developing embedded firmware for battery-health monitoring, autonomous mode transitions, and sensor-data logging while accounting for operation in stratospheric conditions reaching −60°C and ~1.2 kPa

Career Page with Email-Based ATS | *Next.js, FastAPI, SQLModel, Tailwind CSS* June 2025

- Developed a full-stack recruitment platform integrating Next.js, FastAPI, SQLModel, and SQLite across frontend, API, and database layers
- Implemented resume uploads, form validation, and structured candidate-data processing to automate application handling

Intersection Signal Simulator | *VHDL, FPGA (Zybo-Z7), Digital Logic, Assembly, Vivado* January 2024

- Designed and implemented a Moore finite-state machine in VHDL to control traffic-intersection states, including timing and transition logic
- Synthesized and deployed the design to a Zybo-Z7 FPGA using Xilinx Vivado, debugging state transitions and timing behavior through Assembly-level and hardware testing

Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, SQL, VHDL, MATLAB, HTML/CSS, Bash

Software & Web: Agile, React, Next.js, Node.js, FastAPI, Flask, Django, Tailwind CSS, REST APIs, GraphQL, SQL Databases, End-to-End Testing, Frontend, Backend, Git, Github

Embedded & Hardware: STM32, Arduino, Raspberry Pi, Embedded C/C++, Firmware, Sensor Integration, Prototypes, FPGA Development, VHDL, Finite-State Machines, PCB Design, Altium, Circuit Analysis, Schematics, Soldering

Platforms & Tools: Vivado, VS Code, GCP, Azure, LabVIEW, LTSpice, Excel, SolidWorks, AutoCAD, CAD

Communication: Project Manager, Cross-Functional, Communication Skills, Troubleshooting, Technical Support, Attention to Detail